

Inspection Certificate EN 10204 - 3.1

Certificate no. VP-31-2026-1187 | Issue date: 22 August 2026

Manufacturer Vektor Precision Components GmbH	Customer RuhrWorks Manufacturing AG
Purchase order RW-PO-260814-77	Delivery note VPC-DN-88214
Material 1.4404 / X2CrNiMo17-12-2	Declared heat H-26-7718

Product identification

Line	Product	Dimensions	Quantity	Heat / lot
10	Precision valve body blank	48 x 92 mm	240 pcs	H-26-7718 / L-418
20	Precision valve body blank	48 x 92 mm	120 pcs	H-26-771B / L-419

Traceability discrepancy
Line 20 records heat H-26-771B (letter B), while the certificate header and chemical analysis identify heat H-26-7718 (digit 8). The second heat has no independent analysis in this package.

Chemical composition - ladle analysis

Element	Result %	Specification %	Assessment
C	0.018	max. 0.030	Pass
Cr	17.21	16.50 - 18.50	Pass
Ni	9.72	10.00 - 13.00	Below minimum
Mo	2.18	2.00 - 2.50	Pass
Mn	1.42	max. 2.00	Pass

Out-of-specification result
Nickel is reported at 9.72%, below the stated 10.00% minimum for the ordered grade. No concession or material-deviation approval is attached.

Inspection Certificate EN 10204 - 3.1

Mechanical tests, inspection, and declaration

Mechanical test results

Test	Result	Requirement	Assessment
Tensile strength Rm	558 MPa	520 - 670 MPa	Pass
Yield strength Rp0.2	274 MPa	min. 220 MPa	Pass
Elongation A5	43%	min. 40%	Pass
Hardness	178 HBW	max. 215 HBW	Pass
PMI screening	Performed on 18 of 360 pieces	Customer plan: 10%	Insufficient sample

Surface and dimensional inspection

Visual inspection: acceptable. Dimensional report DR-418-26: all sampled dimensions within drawing RW-VB-330 revision C. Sampling level stated as 12 pieces from the delivered quantity of 360 pieces.

Declaration / Erklaerung

We declare that the products listed were manufactured and inspected according to the purchase order, subject to the results and remarks recorded in this certificate. / Wir bestaetigen die Pruefung der aufgefuehrten Erzeugnisse gemaess Bestellung, vorbehaltlich der dokumentierten Ergebnisse und Hinweise.

Authorization incomplete
The inspector name and approval identifier are blank. A company stamp image is present, but no authorized signature or verifiable digital-signature metadata is included.

Inspection representative	Release status
Name: _____	Conditional - quality review required
Approval ID: _____	Stamp: VPC QUALITY CONTROL
Signature: _____	Date: 22 August 2026

Synthetic evaluation note: This fixture deliberately contains a traceability conflict, a chemistry deviation, insufficient PMI sampling, and an unsigned release declaration.